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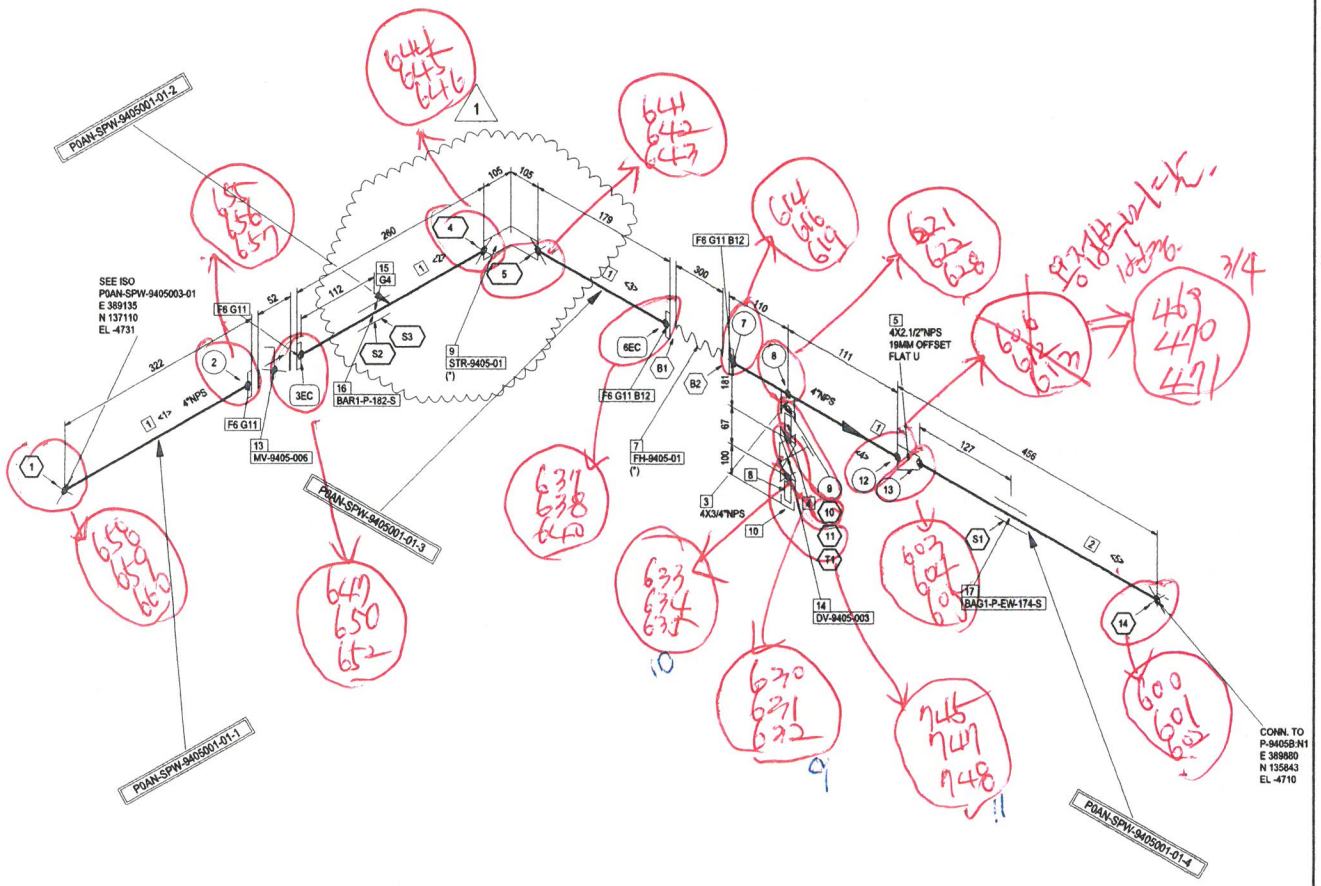

BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	4"	1.0 M	S98	P	PA02-000001904
2	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2.1/2"	0.4 M	S98	P	PA02-000001912
3	HALF COUPLING SW CL3000 ASTM A182-F316/316L	4X3/4"	1	S98	TH	PB99-000038951
4	PIPE NIPPLE PBE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPPG	PB99-000044539
5	ECC. REDUCER BW WELDED ASTM A403-WP316/316L SCH10S * SCH10S	4X2.1/2"	1	S98	RE	PB99-000038601
6	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	4"	4	S98	F	PB01-000023957

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
7	TAG NO : FH-9405-01, FLEXIBLE HOSE	4"	1			SPECIALTY
8	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
9	TAG NO : STR-9405-01, STRAINER - T-TYPE	4"	1			SPECIALTY
10	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
11	SPIRAL WOUND GASKET O-RING: SS 304 CL150 HOOP : SS 316L FILLER : GRAPHITE 4.5MM(0.18") THK.	4"	4	S98	G	PF01-000009269
12	100 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNIU NR x MM	5/8"	16	S98	B	PB99-000076890
13	BUTTERFLY VALVE LUG CL150 ASTM A351-CF8M SS316 TRIM RTFE SEAT SHAFT 17-4 PH LV. OP. (VAY150A)	4"	1	S98	VU	PP07-000003382
14	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443
15	G4	4"	1			G4
16	BAR1-P-182-S	4"	1			BAR-P_P
17	BAG1-P-EW-174-S	2.1/2"	1			BAG-P_P

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	312	4		<2>	250	4	
<3>	169	4		<4>	211	4	
<5>	354	2.1/2					

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-11-11

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION TH'K (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
4	S98	ATM	25 ~ 80	3.5	128	-	S	ISN	25	-	N	5	K1-102
0.75	S98	ATM	25 ~ 80	3.5	128	-	S	ISN	25	-	N	5	K1-102

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
1-A	11-11-25		H.S.JUNG	M.H.SUNG	T.H.LIM
1	14-05-25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	15-04-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P0AN
P&ID NO.	K1-DSL-UT-PID-9405			PROJ. NO.	16480	FLUID	SPW
CAD NO.	-			LINE NO.	4-SPW-9405001-S98-ISN		
DWG NO.	P0AN-SPW-9405001-01			SHT NO.	1 OF 1	REV.	1-A

PIPE modified at 17:03:4 Jun 2025, Drawing Created at 17:05:09 4 June 2025
PIPE NAME : 4-SPW-9405001-S98-ISN_P0AN-01

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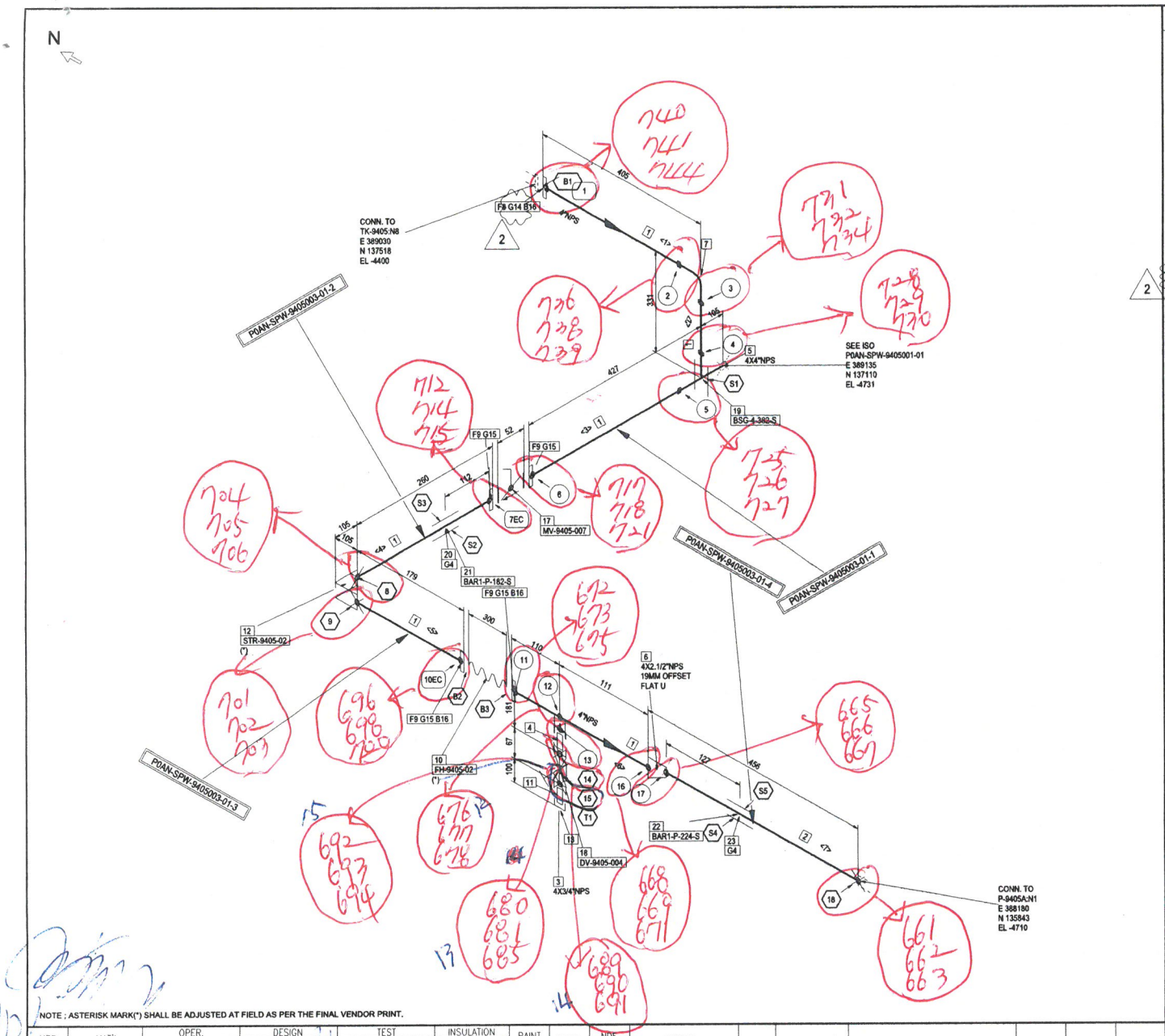

BILL OF MATERIALS

FABRICATION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
1	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	4"	1.3 M	S98	P	PA02-000001904
2	PIPE EFW ASTM A312-316/316L SCH10S BE RT=100%	2.1/2"	0.4 M	S98	P	PA02-000001912
3	HALF COUPLING SW CL3000 ASTM A182-F316/316L	4X3/4"	1	S98	TH	PB99-000038851
4	PIPE NIPPLE PBE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPPG	PB99-000044539
5	TEE BW WELDED ASTM A403-WP316/316L SCH10S	4X4"	1	S98	T	PB99-000038692
6	ECC. REDUCER BW WELDED ASTM A403-WP316/316L SCH10S * SCH10S	4X2.1/2"	1	S98	RE	PB99-000038601
7	90 DEG. ELBOW (LR) BW WELDED ASTM A403-WP316/316L SCH10S	4"	1	S98	E	PB99-000038411
8	SLIP-ON FLANGE FF CL150 ASTM A182-F316/316L	4"	1	S98	FF	PB01-000024050
9	SLIP-ON FLANGE RF CL150 ASTM A182-F316/316L	4"	4	S98	F	PB01-000023852

ERECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT'L CLASS	SHORT CODE	ITEM
10	TAG NO : FH-9405-02, FLEXIBLE HOSE	4"	1			SPECIALTY
11	PIPE NIPPLE TOE(NPT)/POE SMLS ASTM A312-316/316L SCH40S 100MM(4.0")	3/4"	1	S98	NPTG	PB99-000123907
12	TAG NO : STR-9405-02, STRAINER - T-TYPE	4"	1			SPECIALTY
13	CAP SCRD (NPT) CL3000 ASTM A182-F316/316L	3/4"	1	S98	CPT	PB99-000038501
14	SHEET GASKET COMP. NON-ASBESTOS FULL FACE CL150 ARAMID FIBER WITH S BR BINDER 3.0MM(0.12") THK. GARLOK BLUE GARD STYLE 3200	4"	1	S98	GF	PF01-000009408
15	SPIRAL WOUND GASKET O-RING 36 304 CL150 HOOP : SS 316L FILLER : GRAPHIT E 4.5MM(0.18") THK.	4"	1	S98	G	PF01-000092659
16	100 STUD BOLT W/2-NUT HEAVY HEX. NUT ASTM A193-B8 CL.2 ASTM A194-8 UNJ NR x MM	5/8"	24	S98	B	PB99-000076890
17	BUTTERFLY VALVE LUG CL150 ASTM A351-CF8M SS316 TRIM RTFE SEAT SHAFT 1 7-4 PH L.V. OP. (VAY150A)	4"	1	S98	VU	PP07-000003362
18	GATE VALVE SW CL800 ASTM A182-F316/316L SS316 TRIM STELLITE SEATS BB OS &Y SOLID WDG DISC HW. OP (VAV005)	3/4"	1	S98	VA	PP02-000010443
19	BSG-4-382-S	4"	1			BSG_P
20	G4	4"	1			G4
21	BAR1-P-182-S	4"	1			BAR-P_P
22	BAR1-P-224-S	2.1/2"	1			BAR-P_P
23	G4	2"	1			G4

CUT PIPE LENGTH

PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS	PIECE NO	LENGTH (MM)	NPD (IN)	REMARKS
<1>	243	4		<2>	74	4	
<3>	312	4		<4>	250	4	
<5>	189	4		<6>	211	4	
<7>	354	2.1/2					

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2026-01-21

NOTE ; ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (inch)	MAT'L CLASS	OPER. PRESS. (bar.g)	OPER. TEMP. (°c)	DESIGN PRESS. (bar.g)	DESIGN TEMP. (°c)	TEST PRESS. (bar.g)	TEST MEDIUM	INSULATION TYPE	INSULATION THK (mm)	PAINT CODE	PWHT	NDE (RT,%)	STRESS
4	S98	ATM	80	3.5	128	-	S	ISN	25		N	5	K1-102
2.5	S98	ATM	80	3.5	128	-	S	ISN	25		N	5	K1-102
0.75	S98	ATM	80	3.5	128	-	S	ISN	25		N	5	K1-102

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2	21-01-26	ISSUED FOR REVISION	N.H.PARK	M.H.SUNG	T.H.LIM
1	14-06-25	ISSUED FOR REVISION	H.S.JUNG	M.H.SUNG	T.H.LIM
0	15-04-25	ISSUED FOR INFORMATION	H.S.JUNG	M.H.SUNG	T.H.LIM

PROJECT
PROJECT PROVIDENCE

LOTTE BIOLOGICS
LOTTE E&C

PART	AG	PLANT	N	AREA	PDB	CWA NO.	P0AN
P&ID NO.	K1-DSL-UT-PID-9405			PROJ. NO.	16480	FLUID	SPW
GAD NO.	-			LINE NO.	4-SPW-9405003-S98-ISN		
DWG NO.	P0AN-SPW-9405003-01			SH'T NO.	1 OF 1	REV	2

PIPE modified at 15:44:14 Jan 2026 , Drawing Created at 18:46:43 15 January 2026
PIPE NAME : 4-SPW-9405003-S98-ISN_P0AN-01

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